

SPEAKER SCRIPT | 5-7 MINUTES

Stages 4-10 Bilingual Speaker Script

Stage 4-10 双语讲稿

Cash-or-Stock Election Arbitrage

Outcome Overlay through Historical Daily Strategy Results

Use one language version during the presentation. Each version is designed for approximately 6 to 7 minutes.

演示时选择其中一种语言即可；英文版和中文版分别约为 6 至 7 分钟。

Stage 4 | Outcome Overlay

结果状态覆盖层

About 75 seconds / 约 75 秒 | 85.0% completed | 11.3% terminated | 3.8% withdrawn | OOS Brier 0.307

English

Up to this point, Monte Carlo assumes that the deal completes. Stage 4 removes that assumption by assigning every deal three probabilities: completed, terminated, and withdrawn.

The three boxes show average predicted probabilities across our 293-deal deployment universe: 85.0 percent completed, 11.3 percent terminated, and 3.8 percent withdrawn. They are probability weights, not accuracy. Completed paths use the election and proration Monte Carlo. Terminated paths use a target break value with at least a 25 percent loss floor, while withdrawn paths use a more severe 35 percent floor.

The prior does not come from the 293-deal completion-only deployment frame. It comes from 1,997 all-status Bloomberg labels: 1,648 completed, 258 terminated, and 91 withdrawn. We regularize those frequencies with 25 pseudo-observations at the 88, 7, and 5 percent scenario defaults, producing a prior of about 82.6, 12.8, and 4.6 percent. The tuned model then blends 75 percent of that prior with 25 percent of the feature-based Naive Bayes posterior.

The inputs are three numeric features: log announced value, TV-to-EBITDA, and announcement year. The six categorical features are deal type, payment type, target and acquirer market suffixes, and indicators for whether each side has a CUSIP. Deal Status is the training label, not an input, and no future election result is used.

We tune the model inside expanding-year validation. Its temporal out-of-sample Brier score is 0.307, versus 0.317 for a prior-only forecast. That is a modest 3 percent skill gain. Balanced accuracy is 46 percent, so this is not a perfect classifier. Its practical value is that completion is no longer treated as certain.

中文

在这一层之前，Monte Carlo 默认交易一定完成。Stage 4 取消了这个假设，为每笔交易预测三种状态的概率：completed、terminated 和 withdrawn。

底部三个方块是 293 个部署样本的平均预测概率：完成 85.0%，终止 11.3%，撤回 3.8%。这些数字是模拟中的概率权重，不是分类准确率。Completed 路径继续使用 election 和 proration Monte Carlo；terminated 路径使用至少 25% 的 target break-loss floor；withdrawn 路径使用更严格的 35% loss floor。

这里的 prior 不是来自 293 个 completion-only deployment events。它来自 1,997 条包含全部状态的 Bloomberg 训练记录，其中 1,648 条 completed、258 条 terminated、91 条 withdrawn。模型再加入 25 个按照 88%、7%、5% 默认情景分配的伪样本进行收缩，得到约 82.6%、12.8%、4.6% 的 class prior。调优后的最终概率由 75% prior 和 25% feature-based Naive Bayes posterior 混合得到。

分类器使用三个数值特征：log announced value、TV/EBITDA 和 announcement year。六个类别特征是 deal type、payment type、target 和 acquirer 的 market suffix，以及双方是否有 CUSIP。Deal Status 是训练标签，不是输入变量；未来的 election result 也没有进入模型。

模型使用 expanding-year validation 调参，也就是只用更早年份预测之后年份。Temporal out-of-sample Brier score 是 0.307，优于 prior-only 的 0.317，大约有 3% 的概率预测提升。Balanced accuracy 是 46%，所以我们不会把它说成完美分类器。它真正的价值是让后续估值不再假设 completion probability 等于 100%。

Stage 5 | Risk-Adjusted Monte Carlo

风险调整后的 Monte Carlo

About 45 seconds / 约 45 秒 | 32 MC-ready deals | completed-state edge 7.70% | risk-adjusted mean 1.94% | p5 -2.62%

English

Stage 5 inserts the three outcome states into the payoff simulation.

The blue distribution is the completed-state proration edge. Across the 32 MC-ready deals, its mean is 7.70 percent. The purple distribution replaces some paths with terminated and withdrawn losses using the Stage 4 probabilities. After that overlay, the mean falls to 1.94 percent and the fifth percentile is negative 2.62 percent.

The election mechanism creates a completed-deal edge, but close risk consumes a large part of it and creates a left tail. The next stages therefore test whether that risk-adjusted distribution is still tradable.

中文

Stage 5 把 Stage 4 的三种结果状态真正放进 payoff simulation。

图中的蓝色分布只考虑 deal completed 时的 proration edge。32 个 MC-ready deals 的 completed-state 平均 edge 是 7.70%。紫色分布则按照 Stage 4 的概率，把部分路径替换成 terminated 和 withdrawn 的损失。加入 close risk 后，平均收益下降到 1.94%，第五分位数下降到 -2.62%。

这张图最重要的信息是：election mechanism 在 deal 完成时确实能够创造 edge，但 close risk 会消耗其中很大一部分，并形成明显的左尾。因此后续阶段不能只判断哪一边的 consideration 更高，还必须判断风险调整后的分布是否仍然值得交易。

Stage 6 | Direction and Risk Gates

交易方向与风险门槛

About 45 seconds / 约 45 秒 | 88 priced events | 10 ENTER | 10 REVERSE | 20 REVIEW | 48 PASS-family

English

Stage 6 turns each priced payoff distribution into an action.

ENTER means buying the target and exercising the richer election right. REVERSE means shorting the target and accepting the passive blended settlement liability. Both directions must clear the same three checks: expected return above the hurdle, fifth-percentile return above the downside floor, and loss probability below the ceiling.

Out of 88 priced events, 10 become ENTER, 10 become REVERSE, 20 are REVIEW, and 48 are in the PASS family. The large REVIEW and PASS groups are intentional. REVIEW isolates inconsistent deal terms. PASS means the deal is understandable, but its return or downside is not good enough. We preserve uncertainty rather than force every event into a trade.

中文

Stage 6 把每个有价格的 payoff distribution 转化成具体行动。

ENTER 表示买入 target，并使用股东拥有的 richer election right。REVERSE 表示做空 target，并承担 passive blended settlement liability。两个方向都必须通过三道相同的门槛：expected return 高于 hurdle、第五分位收益高于 downside floor、loss probability 低于上限。

88 个 priced events 中，10 个进入 ENTER，10 个进入 REVERSE，20 个进入 REVIEW，48 个属于 PASS family。REVIEW 和 PASS 数量多是设计结果，不是模型失败。REVIEW 用于隔离条款内部不一致或可能误解析的交易；PASS 表示交易结构可以理解，但收益或下行风险不够好。我们的目标是保留不确定性，而不是强迫每笔 event 都产生交易。

Stage 7 | Holder Structure

股东结构

About 40 seconds / 约 40 秒 | 293 estimates | 261 rolling fits | 32 defaults | median p 0.30 | median q 0.10

English

Stage 7 asks who is on the other side of our trade.

The holder model separates passive ownership from active ownership, then splits active holders into noisy and EV-sensitive behavior. The parameter p is the noisy holder's cash-election probability, while q is the EV-sensitive ownership share.

Of 293 estimates, 261 use rolling prior-event fits and 32 use defaults. Median p is 0.30 and median q is 0.10. The 32.1-point demand MAE shows why this is not a standalone demand forecast. It is a capacity prior for share supply, borrow supply, buyer demand, and self-impact.

中文

Stage 7 关注的是：谁会成为我们交易的对手方。

Holder model 先把 passive ownership 和 active ownership 分开，再把 active holders 分成 noisy behavior 和 EV-sensitive behavior。参数 p 表示 noisy holder 选择 cash 的概率；参数 q 表示 EV-sensitive ownership 的比例。

293 个 estimates 中，261 个使用 prior events 的 rolling fit，32 个使用 default。 p 的中位数是 0.30， q 的中位数是 0.10。Demand MAE 达到 32.1 个百分点，所以这个模型不能作为独立的 election-demand forecast。它的定位是 capacity prior，用来估计 share supply、borrow supply、buyer demand 和 self-impact。

Stage 8 | Capacity and Self-Impact

容量与自我影响

About 60 seconds / 约 60 秒 | 20/20 sized | \$621.4m target-leg notional | \$39.9m expected P&L | median position 0.74%

English

Stage 8 converts the 20 risk-approved directions into feasible position sizes.

All 20 candidates receive positive capacity. For ENTER, capacity is limited by available sellers, ten days of trading at no more than 20 percent of ADV per day, a 5 percent position limit, and the possibility that our own election changes proration. For REVERSE, capacity is the minimum of borrowable shares, ADV, the position limit, and noisy-buyer demand. In this run, noisy-buyer demand is the binding constraint for all 10 REVERSE trades.

The 621.4 million dollars is the sum of optimal target-leg notionals across the 20 opportunities. It is not simultaneous capital and excludes the acquirer hedge. Expected P&L is 39.9 million dollars, equivalent to a 6.42 percent notional-weighted expected return.

The median position is 0.74 percent of target shares. The 44.25 percent self-impact threshold is the median among four ENTER deals with a finite threshold. It marks where an extremely large position would fail a risk gate; actual positions are far below it.

中文

Stage 8 把前一层选出的 20 个交易方向转化成实际可行的仓位。

20 个候选交易全部得到正的 capacity。对于 ENTER，容量受到可购买股份、十天建仓且每天不超过 20% ADV、5% position limit，以及我们自己的 election 对 proration 产生影响的限制。对于 REVERSE，容量取 borrowable shares、ADV、position limit 和 noisy-buyer demand 中的最小值。本轮 10 个 REVERSE trades 全部由 noisy-buyer demand 决定最终上限。

621.4 million dollars 是 20 个历史机会的 optimal target-leg notional 总和。它不是同一天需要投入的资本，也不包括 acquirer hedge leg。对应的 expected P&L 是 39.9 million dollars，相当于 6.42% 的 notional-weighted expected return。

Median position 是 target shares outstanding 的 0.74%。44.25% self-impact threshold 只是在四个存在有限 threshold 的 ENTER deals 中取中位数，表示仓位大到什么程度时交易会无法通过风险门槛。实际 optimal positions 远低于这些 threshold。

Stage 9 | Combined Trade Book

组合交易账簿

About 50 seconds / 约 50 秒 | \$39.9m expected P&L | \$17.0m validated realized P&L | expected-realized correlation 0.78

English

Stage 9 aggregates the sized ENTER and REVERSE trades into one strategy view.

The left bar is the 39.9 million dollars of ex-ante expected P&L. Sixteen trades have enough final information for direct realized-payoff validation, producing about 17.0 million dollars of self-impact-adjusted realized P&L. The expected versus realized return correlation is positive 0.78, which suggests that the signal ranks these completed outcomes in a useful direction.

This remains a small, completion-heavy sample. The 6.42 percent is total expected P&L divided by total notional across 20 trades, not a cumulative historical return. The next slide changes the unit from a trade to a daily portfolio.

中文

Stage 9 把完成 sizing 的 ENTER 和 REVERSE trades 汇总成统一的 strategy view。

左侧第一根柱是 39.9 million dollars 的 ex-ante expected P&L。16 笔交易有足够的最终信息，可以直接验证 realized payoff，对应大约 17.0 million dollars 的 self-impact-adjusted realized P&L。Expected return 与 realized return 的相关系数是正 0.78，说明信号在这些 completed outcomes 中具有一定的排序能力。

不过，这仍然是一个较小而且 completion-heavy 的验证样本。这里的 6.42% 是 20 笔交易的 total expected P&L 除以 total notional，不是历史累计收益。下一页会把分析单位从单笔 trade 改成 daily portfolio。

Stage 10 | Historical Daily Strategy Results

历史日度策略结果

About 75 seconds / 约 75 秒 | 18/20 paths | Sharpe 1.08 active / 0.45 calendar | Sortino 3.07 / 1.27 | Max DD -15.21%

English

Stage 10 reconstructs daily target and acquirer hedge P&L for 18 of the 20 selected trades. Fourteen paths reconcile to an observed settlement return, four retain a market-to-deadline fallback, and two trades lack enough common target and acquirer price history.

Each day, concurrent positions are weighted by their optimal notionals. On the 771 days with an active position, mean daily return is 0.13 percent, annualized volatility is 29.33 percent, Sharpe is 1.08, and Sortino is 3.07. When we retain all 4,469 business days and record inactive days as zero return, mean return falls to 0.02 percent, volatility to 12.20 percent, Sharpe to 0.45, and Sortino to 1.27. Maximum additive drawdown is negative 15.21 percent in both versions.

The two charts are different views of the same daily portfolio series. The left attributes return to events in close-date order; the right shows it through calendar time. Both end at 96.67 percent, the additive sum of daily capacity-weighted returns. This exceeds Stage 8's 6.42 percent average expected return because capital is reused across sequential opportunities.

All 18 reconstructed outcomes are completed. The history supports the election and sizing mechanism, but it is not a survivorship-free live-performance estimate. A full backtest still needs realized break paths and execution costs.

中文

Stage 10 为 20 笔 selected trades 中的 18 笔重构 target 和 acquirer hedge 的 daily P&L。14 条路径最终对齐到 observed settlement return，4 条保留 market-to-deadline fallback，另外 2 笔因为 target 和 acquirer 缺少足够的共同价格历史而无法重构。

每天的并发持仓按照 optimal notional 加权。在 771 个有 active position 的交易日中，mean daily return 是 0.13%，annualized volatility 是 29.33%，Sharpe 是 1.08，Sortino 是 3.07。如果保留全部 4,469 个工作日，并把没有持仓的日期记为零收益，mean daily return 下降到 0.02%，volatility 下降到 12.20%，Sharpe 变成 0.45，Sortino 变成 1.27。两种口径的 maximum additive drawdown 都是 -15.21%。

下面两张图来自完全相同的 daily portfolio return series。左图按照 close date 展示每个 event 对累计收益的贡献；右图按照日历时间展示同一条收益路径。两张图最终都达到 96.67%，因为这里展示的是 capacity-weighted daily returns 的 additive sum。它高于 Stage 8 的 6.42% average expected return，是因为资金可以在不同时期的交易机会中重复使用。

最后需要强调限制：18 条可重构路径全部来自 completed outcomes。因此这些结果支持 election mechanism 和 sizing logic，但还不能被解释为没有 survivorship bias 的 live-performance estimate。完整回测仍然需要 terminated 和 withdrawn deals 的真实 break paths，以及 borrow、financing 和 transaction costs。